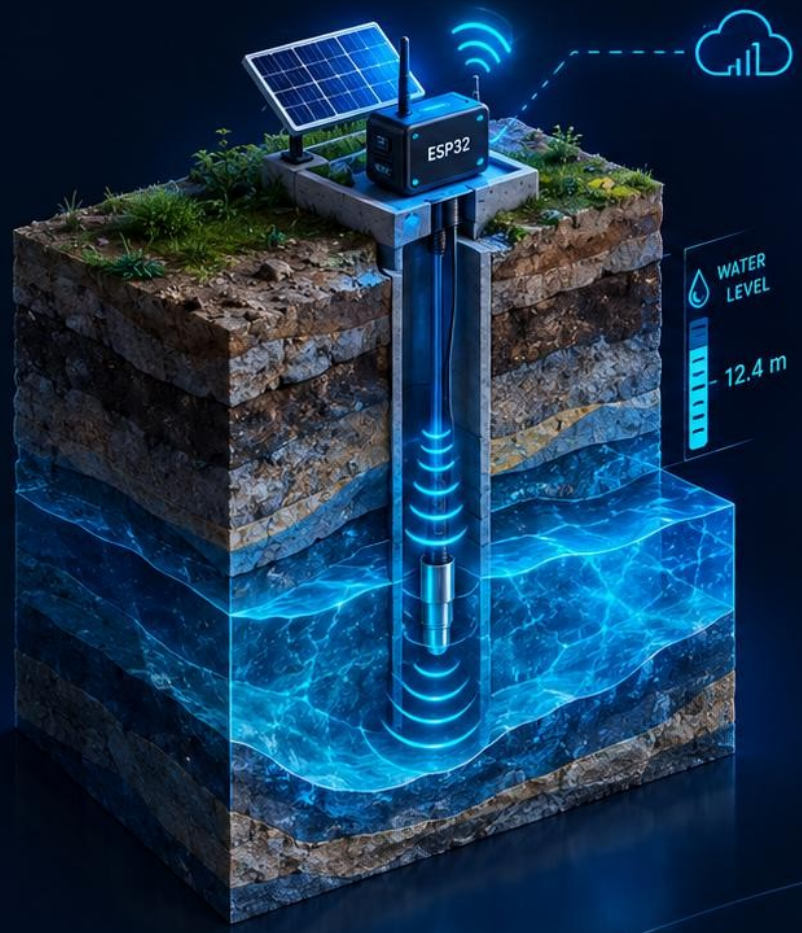
Techfest IIT Bombay | InnovateX



AUTONOMOUS UNDERGROUND WATER MANAGEMENT

Low-Cost Solar IoT Edge Telemetry &
Predictive Depletion Prevention

INDIA'S GROUNDWATER CRISIS

THE INVISIBLE CRISIS:
RUNNING DRY IN THE DARK

25%

World's Largest Groundwater Extractor (CGWB)

>500 PPM

Dangerous Salinity & Silt Ingress (Unfit for Drinking)

0 ALERTS

Pumps Burn Out & Wells Dry Up With Zero Early Warning



DEPLETING AQUIFERS | IMPACTING FARMERS | THREATENING COMMUNITIES | A NATIONAL CHALLENGE

SMARTER MONITORING / HEALTHIER AQUIFERS / SUSTAINABLE TOMORROW

BRIDGING THE CRITICAL MONITORING GAP

FROM MANUAL GUESSWORK ⟶ TO REAL-TIME INTELLIGENCE



Manual Dip Wells

- ✗ Measured 2-4x/year
- ✗ High human error
- ✗ Zero quality data
- ✗ No alerts



Commercial DWLR Loggers

- ₹ >₹50,000 High Cost
- ✗ Level only
- ✗ Vendor locked
- ✗ Not viable for villages



Our Autonomous IoT System

- ✓ <₹3,500 Low Cost
- ✓ Level + TDS + Turbidity
- ✓ Automated Pump Cutoff
- ✓ Instant Telegram Alerts

REAL-TIME WATER DATA | COST-EFFECTIVE DEPLOYMENT | PREVENTS PUMP BURNOUT | BUILDS RESILIENT RURAL COMMUNITIES

BETTER DATA / BETTER DECISIONS / MORE WATER

END-TO-END EDGE & CLOUD ARCHITECTURE

1. SENSING ARRAY



Ultrasonic Level



TDS Probe



DS18B20 Temp



Turbidity

2. DUAL-CORE ESP32 EDGE



Temperature Compensation, Median Filter, Drawdown cm/day Math

3. HARDWARE INTERLOCK



5V Relay
Dry-Run Pump Cutoff
(Active Protection)

4. TELEMETRY & ALERTS



Local Web Dashboard



Telegram Bot Mobile Alert

REAL-TIME DATA | SAFE OPERATION | SMART MONITORING | CLOUD CONNECTED

ENGINEERED FOR FIELD RELIABILITY

ACCURATE / DURABLE / INTELLIGENT

Dual Temperature Compensation

$v = 331.3 + 0.606T$ (Sound)

2%/°C TDS Compensation

T (°C) → $v = 331.3 + 0.606T$

TDS → TDS × 1.02 (per °C)

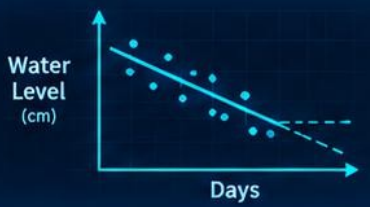
Anti-Electrolysis GPIO Gating

Sensor powered for only **50ms** per reading.
Prevents probe corrosion and extends life **10x**.



Predictive Days-to-Dry Algorithm

On-chip least-squares linear regression on RTC memory predicts days before pump cavitation.



$y = a + bx$

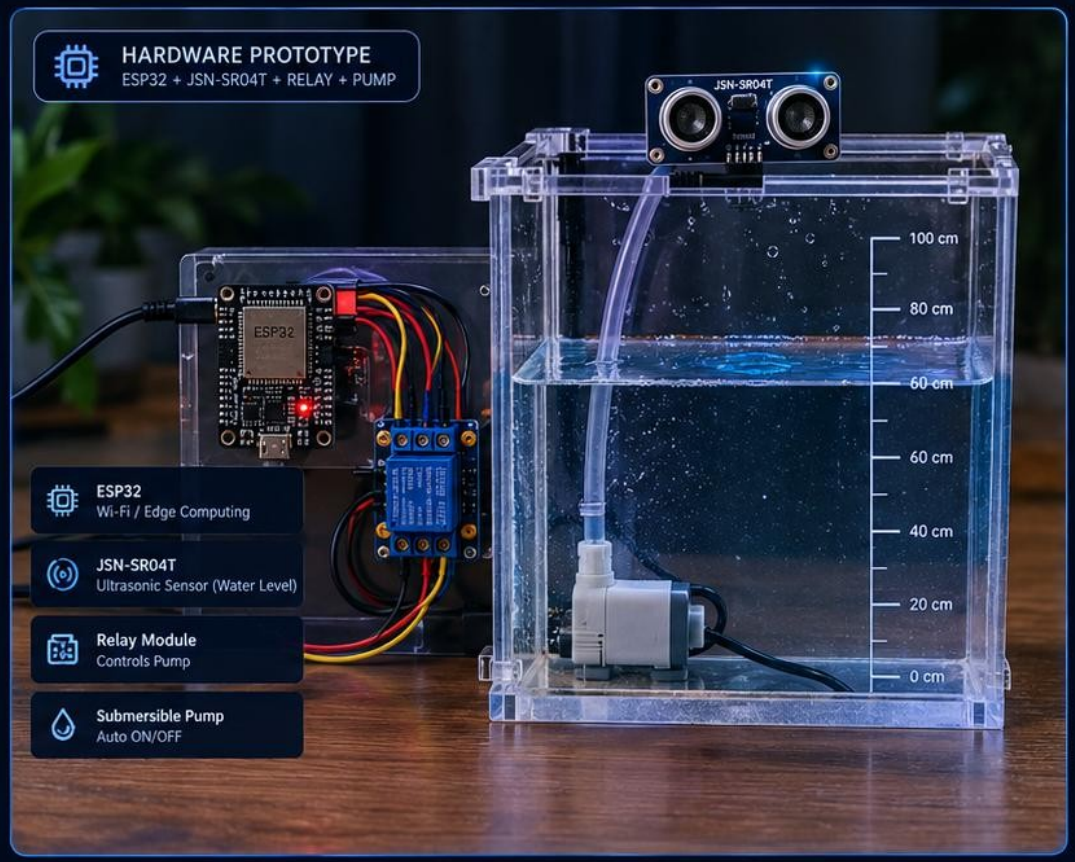
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Days to Dry (Prediction)

RELIABLE SENSORS | SMART EDGE PROCESSING | LONGER LIFESPAN | DATA-DRIVEN INSIGHTS

LIVE PROTOTYPE & EDGE DASHBOARD

REAL-TIME MONITORING • SMART WATER MANAGEMENT • IoT ENABLED





Real-time Monitoring
Live water level & quality data

IoT Enabled
Access from anywhere

Smart Automation
Pump control & safety

Water Conservation
Save water, save tomorrow

Smarter Systems for a Sustainable Future

FRUGAL ECONOMICS & NATIONAL SCALABILITY

LOWER COST / LARGER IMPACT / SUSTAINABLE WATER FUTURE

COST ADVANTAGE

FRUGAL • EFFICIENT • IMPACT DRIVEN

₹3,200
Prototype BOM

<₹1,800
Production Target at Volume

10x Cheaper than Commercial Telemetry Loggers



HIGHER AFFORDABILITY = WIDER IMPACT

3-TIER SCALABILITY ROADMAP

FROM PILOT TO NATIONAL IMPACT

TIER 1

Village Pilot:
5-10 Nodes
(WiFi / Hotspot)

TIER 2

Taluka Scale:
100 Nodes
(LoRaWAN + Gateway)

LoRaWAN

TIER 3

State Integration:
Cloud Database aligned with Atal Bhujal Yojana.

Atal Bhujal Yojana

LOWER DEPLOYMENT COST
More nodes. More coverage.

BETTER WATER MANAGEMENT
Data-driven. Sustainable.

RURAL EMPOWERMENT
Smart villages. Stronger communities.

NATIONAL IMPACT
Scalable. Replicable. At scale.

THE ROADMAP: FROM LAB TO VILLAGE WELLS

BUILD / PILOT / SCALE / IMPACT

1 COMPLETED



Working Hardware, Edge Firmware & Alert System

- ESP32 Edge
- Sensor Array
- Real-time Alerts

2 NEXT 60 DAYS



Field Pilot in 5 Dug Wells with Stilling Tubes

- 5 Wells (Region A)
- Stilling Tube Setup
- Data Validation

3 LONG-TERM



LoRa Mesh Network & Vernacular SMS Alerts for Farmers

- LoRaWAN Mesh
- SMS (Regional Languages)
- Farmer Adoption



> “ You cannot manage what you do not measure.
> **Protecting groundwater before the well runs dry.”**

